BST Physics Curriculum Vol 11 Chapter 5 — D_IV^5 Geometry: Coset + Boundary + Coordinates v0.4 SIGNATURE (refilled per Cal #104)

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Vol 11 Chapter 5 — D_IV^5 Geometry: Coset + Boundary + Coordinates

Chapter motivation

D_IV^5 explicit geometric realization: Cartan 1894 classified bounded HSDs; Hua 1958 (Loo-Keng Hua, *Harmonic Analysis of Functions of Several Complex Variables in the Classical Domains*) gave concrete coordinate realizations for type I-IV classical domains. For D_IV^5: bounded domain in $\mathbb{C}^5$ defined by polynomial inequalities; Shilov boundary structure; Bergman boundary; coset structure SO_0(5,2)/[SO(5) × SO(2)]; symmetric-pair decomposition so(5,2) = (so(5) ⊕ so(2)) ⊕ m with isotropy K = SO(5) × SO(2) + 10-dim coset complement m.

BST substrate-cross-link: explicit geometric realization underlies all BST physics; coset m directions = substrate position + momentum operators (Vol 0 Ch 4 §4.2 v0.5 + T2419 + T2422); Shilov boundary conformal structure = SO_0(3,1) Lorentz embedding (Vol 0 Ch 4 §4.6) = 4D spacetime emergence.

Section 5.0b — Reader-grade 3-level pedagogy

Level 1: D_IV^5 explicit Cartan 1894 + Hua 1958 realization: bounded in $\mathbb{C}^5$ by polynomial inequalities; Shilov boundary; coset SO_0(5,2)/[SO(5) × SO(2)]; symmetric-pair so(5,2) = (so(5) ⊕ so(2)) ⊕ m with 10-dim coset directions.

Level 2 (graduate-mathematician): Hua 1958 explicit realization for D_IV_n: D_IV_n = $\{z \in \mathbb{C}^n : 1 - 2(\bar{z}\cdot z) + |z\cdot z|^2 > 0, |z\cdot z| < 1\}$ where $\bar{z}\cdot z = \Sigma\, |z\_k|^2$ (Hermitian inner product) + $z\cdot z = \Sigma\, z\_k^2$ (bilinear). For D_IV^5 ($n = 5$): bounded domain in $\mathbb{C}^5$ via these two polynomial inequalities. Boundary structure: topological boundary ∂D_IV^5 = closure minus interior; Shilov boundary = smallest closed subset on which holomorphic functions attain maximum modulus (Bergman 1947 + Bergman boundary smaller subset). For type IV: Shilov boundary ≅ S^{n-1} × S^1 (sphere bundle); for D_IV^5: Shilov ≅ $S^4 \times S^1$. Conformal structure: SO_0(5,2) acts as conformal transformations on Shilov boundary; SO_0(3,1) × SO(2) ⊂ SO_0(5,2) embedding gives 4D Lorentz + U(1) substructure (Vol 0 Ch 4 §4.6). Substrate 4D spacetime emergence at Shilov boundary: Vol 1 Ch 2 + Vol 5 Ch 1 pedagogical bridge. Symmetric-pair decomposition: so(5,2) = (so(5) ⊕ so(2)) ⊕ m with dim so(5,2) = 21; dim K = SO(5) × SO(2) = 10 + 1 = 11; dim m = 10 = 21 − 11. The 10-dim coset complement m contains substrate translation generators (Vol 0 Ch 4 §4.2 v0.5): 5 position M_z + 5 momentum P_z directions; T2419 + T2422 substrate position + momentum operators live in m (NOT in isotropy SO(5)). Hua coordinate realization underlies all substrate-cartography: every BST observable expressible as evaluation on Hua coordinates of D_IV^5; Bergman kernel K_B(z, w̄) = c_FK · h(z, w̄)^(−7/2) in Hua coordinates (Vol 11 Ch 2). Lie-algebra-level + group-level both essential: K-types V_(p,q) (Vol 11 Ch 3) live in Bergman space via group-level Wallach 1976; substrate operators (Vol 1 Ch 6 zoo) live in m via Lie-algebra-level Cartan decomposition.

Level 3 (5th-grader accessible): D_IV^5 is a specific bounded 5-complex-dimensional space (Cartan 1894 classified; Hua 1958 gave coordinates). It's bounded by two polynomial inequalities. Its boundary has special structure (Shilov boundary = sphere × circle for D_IV^5). The substrate D_IV^5 has a "coset" structure: take the big symmetry group SO_0(5,2) and divide by the "isotropy" SO(5) × SO(2); what's left is the 10-dimensional substrate "translation directions". BST's position + momentum operators live in these translation directions (Vol 0 Ch 4 §4.2 v0.5 fix).

Section 5.1 — Hua 1958 Coordinate Realization

$D\_IV\_n = \{z \in \mathbb{C}^n : 1 - 2(\bar{z}\cdot z) + |z\cdot z|^2 > 0, |z\cdot z| < 1\}$

with $\bar{z}\cdot z = \Sigma |z\_k|^2$ (Hermitian) + $z\cdot z = \Sigma z\_k^2$ (bilinear).

For D_IV^5: bounded in $\mathbb{C}^5$ by these two polynomial inequalities.

Section 5.2 — Shilov + Bergman Boundary

Topological boundary ∂D_IV^5 = closure minus interior.

Shilov boundary = smallest closed subset on which holomorphic functions attain max modulus (Bergman 1947).

For type IV: Shilov ≅ S^{n-1} × S^1; for D_IV^5: Shilov ≅ $S^4 \times S^1$.

Section 5.3 — Conformal Structure + Lorentz Embedding

SO_0(5,2) acts as conformal transformations on Shilov boundary.

SO_0(3,1) × SO(2) ⊂ SO_0(5,2) embedding gives 4D Lorentz + U(1) substructure (Vol 0 Ch 4 §4.6).

Substrate 4D spacetime emergence at Shilov boundary: Vol 1 Ch 2 + Vol 5 Ch 1 pedagogical bridge.

Section 5.4 — Symmetric-Pair Decomposition

so(5,2) = (so(5) ⊕ so(2)) ⊕ m with: - dim so(5,2) = 21 - dim K = SO(5) × SO(2) = 10 + 1 = 11 - dim m = 10 = 21 − 11

10-dim coset complement m contains substrate translation generators.

Section 5.5 — Substrate Position + Momentum in Coset m

Per Vol 0 Ch 4 §4.2 v0.5 fix: substrate position M_z + momentum P_z (T2419 + T2422) live in m (coset translation directions), NOT in isotropy SO(5).

5 position M_z (z_1, …, z_5 multiplication) + 5 momentum P_z (∂*{z_1}, …, ∂*{z_5} Wirtinger) = 10 generators of m.

Angular momentum L = 10 SO(5) rotation generators in isotropy SO(5).

Section 5.6 — All BST Observables in Hua Coordinates

Every BST observable expressible as evaluation on Hua coordinates of D_IV^5; Bergman kernel $K_B(z, \bar{w}) = c_{FK} \cdot h(z, \bar{w})^{(-7/2)}$ in Hua coordinates (Vol 11 Ch 2).

Group-level + Lie-algebra-level both essential: K-types $V_{(p,q)}$ (Vol 11 Ch 3) live in Bergman space via group-level Wallach 1976; substrate operators (Vol 1 Ch 6 zoo) live in m via Lie-algebra-level Cartan decomposition.

Section 5.7 — Honest scope + Connection

- Hua 1958 Cartan 1894 explicit realization ?
- Shilov boundary + 4D Lorentz embedding (Vol 0 Ch 4 §4.6) ?
- Symmetric-pair decomposition + coset m for substrate operators
- Open scope: explicit Hua-coordinate computation tables for BST observables (multi-week)

Connection: - Vol 0 Ch 1 + Ch 4 (D_IV^5 APG + isotropy structure) - Vol 1 Ch 2 Substrate Hilbert Space + Vol 5 Ch 1 pedagogical bridge - Vol 11 Ch 1 + Ch 2 + Ch 3 (Bounded HSDs + Bergman + Wallach) - Vol 0 Ch 4 §4.2 v0.5 coset Cartan decomposition (V1-V4 Saturday fix)

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